Master Experiment Execution Guide

KV-Cache Continuity and Intermediate Representation Limits in Depth-Truncated Transformers

Reviewer Revisions | *Missing Baselines* | *Rigor Diagnostics* | *Speculative Decoding*

Target: Kaggle (P100/T4) | *Colab* | *Local CUDA/CPU*

0. Four Critical Rules (Read Before Spending GPU Hours)

Joint Condition Needs Matching Adapters (Fix 1)

Never evaluate the joint condition using an adapter trained at a single depth (L=18) or on an out-of-domain dataset (TinyStories). Train the multi-depth in-domain adapter covering all candidate depths L in {18, 22, 24, 30} on WikiText-2 first (Phase 3), then evaluate the joint condition at those identical depths.

Calibrated Hypothesis Phrasing (Fix 2)

If the multi-depth adapter reaches an empirical plateau near ~82% Top-1 agreement, do NOT claim an "intrinsic manifold limitation". Instead, report it precisely as: "An empirical representation ceiling for this specific adapter family (bottleneck MLP with residual connection) under the given parameter budget (r=512), rather than an absolute theoretical bound."

Strict Precision & Dtype Consistency (Fix 3)

NVIDIA Tesla P100 GPUs have NO native bfloat16 hardware support, and T4 GPUs emulate bfloat16 slowly. Always pass --dtype float16 across all scripts on Kaggle. NEVER mix dtypes within the same table. Every row in a comparison table must share identical precision.

Gated Llama-3.2 Authentication (Fix 4)

meta-llama/Llama-3.2-1B is a gated Hugging Face repository. Add your HF access token as a Kaggle Secret named HF_TOKEN and ensure Notebook Settings -> Internet is turned ON.

1. Environment Setup & Pre-Flight Smoke Checks

1.1. Kaggle / Colab Initial Setup

```
# Verify GPU accelerator
!nvidia-smi

# Install required dependencies
!pip install -q torch transformers datasets accelerate sentencepiece

# Authenticate Hugging Face for Llama-3.2
import os
from huggingface_hub import login
if "HF_TOKEN" in os.environ:
    login(token=os.environ["HF_TOKEN"])
    print("[OK] Hugging Face authenticated successfully.")
else:
    print("[!] Warning: HF_TOKEN secret not found.")
```

Set repository directory and run unit tests:

```
%cd /kaggle/working/kv_continuity
export PYTHONPATH=.
!python -m unittest discover tests
```

1.2. Pre-Flight Smoke Test Suite (< 3 minutes)

```
# Smoke test Table 1 headline script
python experiments/run_tier1_interventions.py --model-type qwen \
  --model-id Qwen/Qwen2.5-3B --dtype float16 --num-samples 2 \
  --seeds 42 --sweep-depths 18 --skip-swap --skip-cosine \
  --device cuda --output-dir results/smoke_test

# Smoke test Llama shallow sweep
python experiments/run_multiseed_llama_sweep.py \
  --model-id meta-llama/Llama-3.2-1B --depths 4 --seeds 42 \
  --num-samples 2 --device cuda --output-dir results/smoke_test

# Smoke test Causal KV swap
python experiments/run_robust_kv_swap_and_metrics.py \
  --model-id Qwen/Qwen2.5-3B --depths 18 --num-samples 2 \
  --device cuda --output-dir results/smoke_test

# Smoke test Autoregressive generation
python experiments/run_autoregressive_generation.py \
  --model-id Qwen/Qwen2.5-3B --depths 18 --num-samples 2 \
  --max-new-tokens 16 --device cuda --output-dir results/smoke_test

# Smoke test Extra fill baselines
python experiments/run_kv_baselines.py \
  --model-id Qwen/Qwen2.5-3B --depths 18 --num-samples 2 \
  --device cuda --output-dir results/smoke_test
```

2. Priority Phase 1: Core Reviewer Objections

2.1. Standardized Table I (N=30, 3 Seeds) with Cache Omission (A1)

- Goal: Standardizes headline eval on WikiText-2 across N=30 sequences with 3 seeds (42, 123, 7).
- Runtime: ~12-15 mins on T4/P100; ~4 mins on A100.

```
python experiments/run_tier1_interventions.py \
  --model-type qwen --model-id Qwen/Qwen2.5-3B --dtype float16 \
  --num-samples 30 --seeds 42,123,7 --sweep-depths 18,24,30 \
  --skip-swap --skip-cosine --device cuda \
  --output-dir results/canonical_sweeps/revision_table1
```

2.4. Shallow Llama-3.2-1B Depth Sweep (A5)

- Goal: Sweeps shallow exit depths on Llama-3.2-1B across 3 seeds to pinpoint crossing point below ln(2) ~ 0.6931 nats.

```
python experiments/run_multiseed_llama_sweep.py \
  --model-id meta-llama/Llama-3.2-1B --depths 4,5,6,7,8,10,12,14 \
  --seeds 42,123,7 --num-samples 30 --device cuda \
  --output-dir results/canonical_sweeps/revision_llama_shallow
```

2.3. Centered Cosine Similarity & Direct Attn-Output Error (A10)

```
python experiments/run_tier1_interventions.py \
  --model-type qwen --model-id Qwen/Qwen2.5-3B --dtype float16 \
  --swap-depths 18 --num-samples 30 --skip-table1 --skip-swap \
  --device cuda --output-dir results/canonical_sweeps/revision_cosine_attn
```

2.2. Expanded Causal Key/Value Swap Across Depths (A9)

```
# Qwen2.5-3B (Depths 14, 18, 22, 24, 30):
python experiments/run_robust_kv_swap_and_metrics.py \
  --model-id Qwen/Qwen2.5-3B --depths 14,18,22,24,30 \
  --num-samples 30 --device cuda \
  --output-dir results/canonical_sweeps/revision_kv_swap_qwen

# Llama-3.2-1B (Depths 6, 8, 10):
python experiments/run_robust_kv_swap_and_metrics.py \
  --model-id meta-llama/Llama-3.2-1B --depths 6,8,10 \
  --num-samples 30 --device cuda \
  --output-dir results/canonical_sweeps/revision_kv_swap_llama
```

3. Priority Phase 2: Autoregressive Generation & Extra Baselines

4.1. Autoregressive (Non-Teacher-Forced) Generation (A3)

- Goal: Measures compounding error in real open-loop AR generation over 256 tokens.

```
python experiments/run_autoregressive_generation.py \
  --model-id Qwen/Qwen2.5-3B --depths 18,24,30 --num-samples 30 \
  --max-new-tokens 256 --device cuda \
  --output-dir results/canonical_sweeps/revision_autoregressive
```

4.2. Extra Fill Baselines (A2, A6)

```
python experiments/run_kv_baselines.py \
  --model-id Qwen/Qwen2.5-3B --depths 18,24,30 --num-samples 30 \
  --device cuda --output-dir results/canonical_sweeps/revision_kv_baselines
```

4. Priority Phase 3: Multi-Depth Adapter Training & Joint Condition

3.2. Multi-Depth In-Domain Adapter Training (A12)

- Capacity Test: Run 10,000 steps first (~10M tokens). Continue to 25k if GPU hours allow.
- Runtime (10k steps): ~2.5 hours on T4/P100; ~45 mins on A100.

```
python experiments/train_exit_adapters.py \
  --model-type qwen --model-id Qwen/Qwen2.5-3B \
  --candidate-depths 18,22,24,30 --adapter-type mlp --rank 512 \
  --steps 10000 --batch-size 8 --seq-len 128 --eval-every 500 \
  --device cuda --output-dir checkpoints/multi_depth_r512
```

3.1. Joint Early-Exit Condition Evaluation (A4)

```
python experiments/run_joint_condition_sweep.py \
  --model-id Qwen/Qwen2.5-3B --depths 18,22,24,30 \
  --adapter-path checkpoints/multi_depth_r512/final_exit_adapters_qwen_mlp_r512.pt \
  --num-samples 30 --device cuda \
  --output-dir results/canonical_sweeps/revision_joint_condition
```

5. Priority Phase 4: Extended Generalization, Robustness & Downstream

5.1. Long-Context Scaling (T in {1024, 2048, 4096}) (A8)

```
python experiments/run_long_context_scaling.py \
  --model-id Qwen/Qwen2.5-3B --depths 18,24,30 --device cuda \
  --output-dir results/canonical_sweeps/revision_long_context
```

5.2. Downstream Zero-Shot Benchmarks (A7)

```
python experiments/run_downstream_benchmarks.py \
  --model-id Qwen/Qwen2.5-3B --depths 22,24,30 --device cuda \
  --output-dir results/canonical_sweeps/revision_downstream
```

5.3. Entropy & Domain Breakdown (A11)

```
python experiments/run_entropy_domain_comparison.py \
  --model-id Qwen/Qwen2.5-3B --depths 18,24 --device cuda \
  --output-dir results/canonical_sweeps/revision_entropy
```

5.4. Large Model Scaling (7B-8B with Quantization) (A13)

```
python experiments/run_large_model_sweep.py \
  --model-id Qwen/Qwen2.5-7B --depths 14,20,24 --load-in-8bit \
  --device cuda --output-dir results/canonical_sweeps/revision_7b
```

5.5. Active Speculative Decoding with Cascading Reads (A14)

```
python experiments/run_active_speculative_decoding.py \
  --model-id Qwen/Qwen2.5-3B --d1 18 --d2 24 --device cuda \
  --output-dir results/canonical_sweeps/revision_active_spec
```

6. Priority Phase 5: Scheduled Speculative Decoding & Tuned-Lens

6.1. Multi-Token Speculative Smoke Benchmark (gamma=2)

```
python experiments/benchmark_speculative_decoding.py \
  --model-id Qwen/Qwen2.5-3B \
  --adapter-path checkpoints/exit_adapters_qwen_mlp.pt \
  --draft-depth 24 --gamma 2 --max-tokens 40 --dtype float16 --device cuda
```

6.2. Fast Tuned-Lens Calibration Training (< 3 Mins)

Exploits Key-stable / Value-evolving asymmetry via ultra-lightweight rank-64 low-rank affine tuned lens.

```
python experiments/train_tuned_lens.py \
  --model-id Qwen/Qwen2.5-3B --target-depths 14,18,22,24,28 \
  --adapter-type lora --rank 64 --steps 400 --batch-size 2 \
  --seq-len 128 --lr 1e-3 --distill-temp 2.0 \
  --output-path checkpoints/exit_adapters_qwen_tuned_lens.pt --device cuda
```

6.3. Systematic Speculative Depth Sweep

```
# Evaluate baseline MLP adapter:
python experiments/sweep_speculative_depths.py \
  --model-id Qwen/Qwen2.5-3B \
  --adapter-path checkpoints/exit_adapters_qwen_mlp.pt \
  --depths 14,18,22,24,28,30 --gamma 1 --num-prompts 12 \
  --max-tokens 30 --dtype float16 --device cuda \
  --output-dir results/canonical_sweeps/qwen2.5_3b

# Evaluate newly calibrated Tuned-Lens adapter:
python experiments/sweep_speculative_depths.py \
  --model-id Qwen/Qwen2.5-3B \
  --adapter-path checkpoints/exit_adapters_qwen_tuned_lens.pt \
  --depths 14,18,22,24,28,30 --gamma 1 --num-prompts 12 \
  --max-tokens 30 --dtype float16 --device cuda \
  --output-dir results/canonical_sweeps/qwen2.5_3b
```

6.4. Publication-Scale 60-Prompt Multi-Domain Benchmark

```
python experiments/benchmark_large_scale.py \
  --model-id Qwen/Qwen2.5-3B \
  --adapter-path checkpoints/exit_adapters_qwen_tuned_lens.pt \
  --draft-depth 24 --gamma 1 --num-prompts-per-domain 15 \
  --max-tokens 30 --device cuda
```

7. Wall-Clock Runtime Summary

Phase	Description	Est. Time (T4/P100)	Est. Time (A100)
Pre-Flight	Unit Tests & Smoke Tests	~2 - 3 mins	~1 min
Phase 1	Table 1, Llama, Cosine, Swap	~45 - 55 mins	~15 - 18 mins
Phase 2	AR Gen & Extra Fill Baselines	~25 - 32 mins	~8 - 10 mins
Phase 3	Adapter Train (10k) & Joint	~2.5 - 3.0 hrs	~45 - 55 mins
Phase 4	Long-Ctx, Downstream, 7B, Spec	~60 - 75 mins	~20 - 25 mins
Phase 5	Spec Decode & Tuned-Lens Track	~25 - 30 mins	~8 - 10 mins
TOTAL	Entire KV Paper Evidence Suite	~5.0 - 6.0 hrs	~1.8 - 2.2 hrs

7.2. Recommended Two-Session Kaggle Plan

Session 1: All Inference & Evaluation Experiments (~1.5 to 2 hours). Execute Phases 1, 2, 4, and 5. Zip and download results.

```
import shutil
shutil.make_archive("kv_inference_results", "zip", "results")
print("[OK] Session 1 results ready for download!")
```

Session 2: Adapter Training & Joint Condition (~2.5 to 3 hours). Execute Phase 3. Zip and download checkpoints.

```
import shutil
shutil.make_archive("kv_adapter_checkpoints", "zip", "checkpoints")
shutil.make_archive("kv_joint_results", "zip", "results")
print("[OK] Session 2 checkpoints and results ready!")
```